

ME396 Final Lab Project

Background Information:

Overall performance of an engine can be obtained from the proper design of exhaust systems. Due to this, our customer wishes to test their design for an exhaust manifold made of 316 Stainless Steel. They wish to determine if the design is sufficient, in the sense that it will not fail due to thermal load or the flow being exerted inside. They also wish to identify any bottlenecks/inefficiencies in the flow through the manifold. The exhaust manifold is mounted on the cylinder head of the engine on one end, and is connected to the flange for the catalytic converter on the other end. There are bolts connecting the manifold to both the engine's head (on one end) and the catalytic converter flange (on the other end). There may or may not be technologies involved regarding the catalytic converter end of the exhaust, so you should investigate the appropriate boundary condition on this end (in structural analysis).

Specifications:

- The exhaust manifold is designed to accommodate a maximum flow rate of 0.12kg/s of exhaust gas on each of the three ports.
- The backpressure at the outlet of the manifold is 27kPa.
- The engine's exhaust gas exits at approximately 525K (it enters the three port inlets that are attached to the cylinder head, and this is what the fluid temperature should be set in Fluent).
- The exhaust manifold is made of 316 stainless steel (you may need to edit the material to add properties).
- The under-hood temperature of the vehicle is approximately 398K.
- The exhaust manifold undergoes heat transfer due to convection, conduction, and radiation.
 - The convective heat transfer coefficient is modeled as stagnant air across the outer surface of the exhaust manifold with a film coefficient of 10 W/m²*C.
 - The surface has an emissivity of 0.5.
 - The engine in contact with the exhaust manifold's flange (3 ports side) is at 363K.
 - The exhaust flange side (single port) is at 295K (corresponding to standard room temperature).
- Assume the exhaust gas is primarily CO₂ (you must set the fluid in Fluent to CO₂ as opposed to air).

Goals:

- Show the results for the current design.
 - Show the velocity streamlines from Fluent
 - Show the temperature isosurfaces from Fluent
 - Show the temperature distribution from Steady State Heat Transfer
 - Show the von Mises stress from Static Structural
 - Show the safety factor from Static Structural
 - Show the deformation from Static Structural
- Is the customer's design sufficient at these conditions? Show data.
- How important are the thermal losses due to convection/radiation under the hood? Show data. Can radiation and convection be neglected in this simulation? Where are the major differences in the design? What region of the exhaust manifold is affected most?
- Are any changes necessary to the design? Give recommendations, show changes you can implement. Test these changes if possible.
- How do the under-hood conditions affect the design? Show data. Can altering these conditions improve the design's performance?
- How was the catalytic converter end of the exhaust manifold modeled (boundary conditions). Are these realistic?

ME396 Final Lab Project

- Do they affect the solution/failure?
- What is the appropriate boundary condition on this end?
- How did you apply the structural boundary conditions on the outlet side (catalytic converter side) of the exhaust manifold?
- Is this a valid assumption?
- How are exhaust manifolds typically connected at this end (you may need to investigate the literature to determine this).
- Can you identify any areas in the exhaust manifold that would create performance issues? For example, regions of high turbulence, pressure drop, pressure increase, pressure imbalance, etc? Describe them. Is the backpressure between cylinders balanced, and if not, describe a method of improving balance.
- Is the design safe? Do you anticipate failure? If failure were to occur, where would it fail? What failure mode do you expect from this material/design?

Hints:

- Run the simulation with and without the convection/radiation heat transfer (steady state thermal).
- Ensure that you are importing your thermal loads AND your fluid pressure onto the interior of the exhaust manifold in static structural.
- Fluent mesh must be under 50,000 elements/nodes to solve.
- You must use inflation in the Fluent mesh (7 layers are recommended to get a converged solution)
- Thermal/structural meshes must be under 32,000 elements/nodes to solve. Make sure to use more than the default meshing, or your answer will be incorrect. Easiest way to do this is by increasing “resolution” from 2 to a larger number (below the limit).
- The exhaust gas should be modeled as CO₂.
- You may need to add/lookup properties to add to the material.
- The turbulence model should be K-epsilon (realizable).
- Use enhanced wall functions for the K-epsilon model and include viscous heating and thermal effects.
- Turbulence specification should be 5% intensity with a viscosity ratio of 10.
- Solution method for fluent should be COUPLED.
- Make sure fluent is given enough iterations to converge to at minimum 10^{-3}
- You need to determine where the manifold has supports (base this on how it attaches to the engine block/rest of the exhaust). Different types of supports are possible (for example sliding) if you can explain how they can be implemented in the design.
- 316 Steel has a compressive yield strength of 30,000 PSI, and a tensile yield strength of 75,000 PSI (if it not defined in your version of ANSYS, you will need to edit the material and specify this).
- You need to determine which ANSYS modules and solvers are necessary in your project schematic to fully carry out these simulations.
- It's recommended you look into how current exhaust manifolds/exhaust systems are attached to vehicles to determine if any technology exists that could improve your results.

Project Submission:

- The project should be submitted in the form of a full technical report, consisting of introduction, method, results, and conclusion sections.

ME396 Final Lab Project

- You must use an engineering format, for example one of the formats shown on the MIE Labs page on blackboard.
- There should be sufficient figures and explanation of the entire simulation and results.
- Analysis files are to be submitted to blackboard (they will be checked for cheating).
- Reports are to be submitted on Blackboard through the assignment submission link (they will be checked for plagiarism).

Report Format:

- Introduction: This section should explain the problems being simulated and the design goals we wish to achieve. It should introduce the reader to the stated problems.
- Problem Statement: State the problems being simulated. Give the boundary conditions, initial conditions, material properties, mesh attributes, assumptions, and metrics. Show how the solver modules are connected in FSI.
- Method: Section should show your analysis of the problem. Describe your methodology to achieve a suitable result.
- Results: Show plots and contours of your results that achieve your stated goals. Each figure must be captioned, referenced, and explained clearly. Your figure/explanation should convince the reader that you have achieved your design goals.
- Conclusion: Should summarize the results achieved, recommendations made to the customer, and any other important summarized information.

All sections/data need not be present in all reports, however any section/data necessary to reproduce your results must be present.

For detailed information of how to write a proper lab/technical report, refer to:

J. Donnell, S. Jeter, C. MacDougall, and J. Snedeker. *Writing Style and Standards in Undergraduate Reports*. College Publishing, VA, 3rd Ed. 2016.